Telecom Bill Generation System

A Minor Project Submitted to

Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal

Towards Partial Fulfillment for the Award of

Bachelor of Technology

(Computer Science and Engineering)

Under the Supervision of

Prof. Aaquib Multani

Submitted By

Rahul Chauhan

(0827CS231207)

Department of Computer Science and Engineering

Acropolis Institute of Technology & Research, Indore

July-Dec 2025

EXAMINER APPROVAL

The Minor Project entitled "**Telecom Bill Generation System**" submitted by **Rahul Chauhan (0827CS231207)** has been examined and is here by approved towards partial fulfillment for the award of Bachelor of Technology degree in Computer Science Engineering discipline, for which it has been submitted. It understood that by this approval the undersigned do not necessarily endorse or approve any statement made, opinion expressed, or conclusion drawn therein, but approve the project only for the purpose for which it has been submitted.

(Internal Examiner) **(External Examiner)**

Date: **Date:**

RECOMMENDATION

This is to certify that the work embodied in this Minor project entitled "**Telecom Bill Generation System**" submitted by **Rahul Chauhan** (0827CS231207) is a satisfactory account of the bonafide work done under the supervision of **Prof. Aaquib Multani**, is recommended towards partial fulfillment for the award of the Bachelor of Technology (Computer Science Engineering) degree by Rajiv Gandhi Proudyogiki Vishwavidhyalaya, Bhopal.

(Project Guide) **(Project Coordinator)**

REPORT

Telecom Bill Generation System Web Application

1. Introduction

The Telecom Bill Generation System is a Python and MySQL-based application designed to automate customer billing. The system stores customer details, calculates billing amounts based on the number of calls made, applies GST, and displays the final payable amount.

It provides two interfaces:

- Command Line
- Tkinter-based GUI

This system eliminates manual work, reduces human errors, and improves billing efficiency.

2. Objectives of the Project

The main objectives of this project are:

- To automate telecom bill calculation.
- To provide an easy-to-use GUI for billing operators.
- To store customer records securely in a database.
- To generate accurate and error-free bills.
- To implement CRUD operations (Create, Read, Update, Delete).
- To ensure persistent storage using MySQL.

3. System Features

3.1 Customer Management

- Add new customer
- Edit customer details
- Delete customer
- Search customer
- View all stored customer records

3.2 Billing Features

- Automatic calculation of bill based on call usage
- GST calculation (10%)
- Printable formatted bill

4. System Architecture

The system follows a **three-tier architecture**:

Layer	Technology Used
Frontend	Python Tkinter GUI
Backend Logic	Python
Database	MySQL

5. System Requirements

5.1 Software Requirements

- Python 3+
- MySQL Server & Workbench
- mysql-connector library
- Tkinter module (preinstalled)

5.2 Hardware Requirements

- 4GB RAM
- Core i3 or above
- 10GB Free Disk Space

6. Implementation Details

Database Table

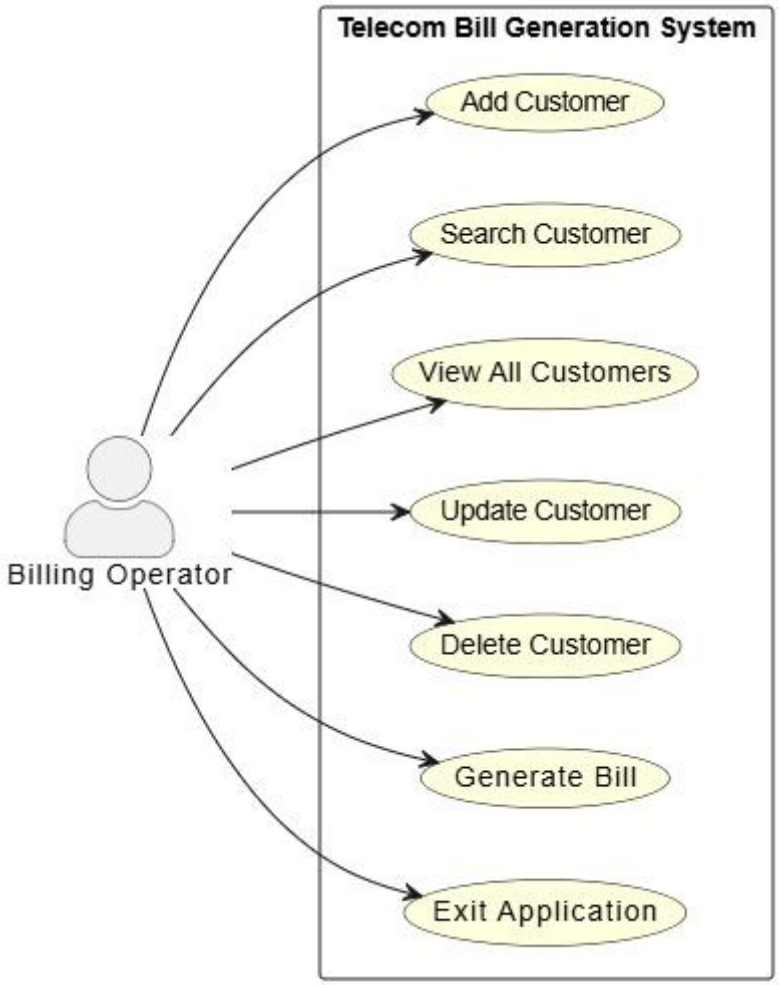
Field Name	Type
cno	INT (PK)
cname	VARCHAR(50)
mobile	BIGINT
address	VARCHAR(100)
email	VARCHAR(100)
no_calls	INT

7. Billing Formula

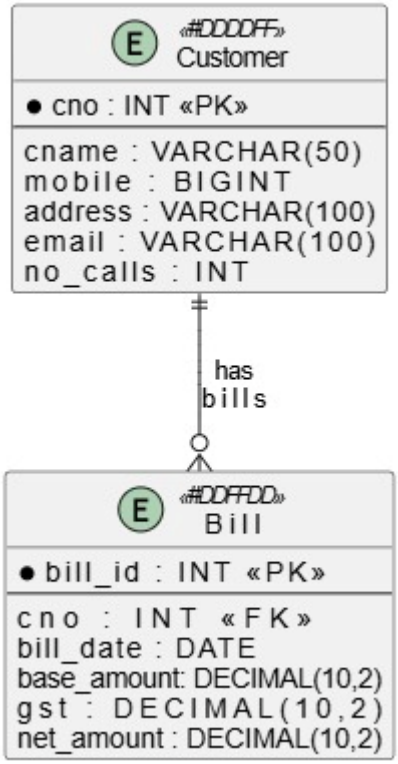
1. If calls ≤ 100 → Amount = ₹200
2. If 101–150 → ₹200 + ₹0.60 × (calls − 100)
3. If 151–200 → previous + ₹0.50 × (calls − 150)
4. If > 200 → previous + ₹0.40 × (calls − 200)
5. GST = 10%
6. Net Bill = Amount + GST

8. UML Diagrams

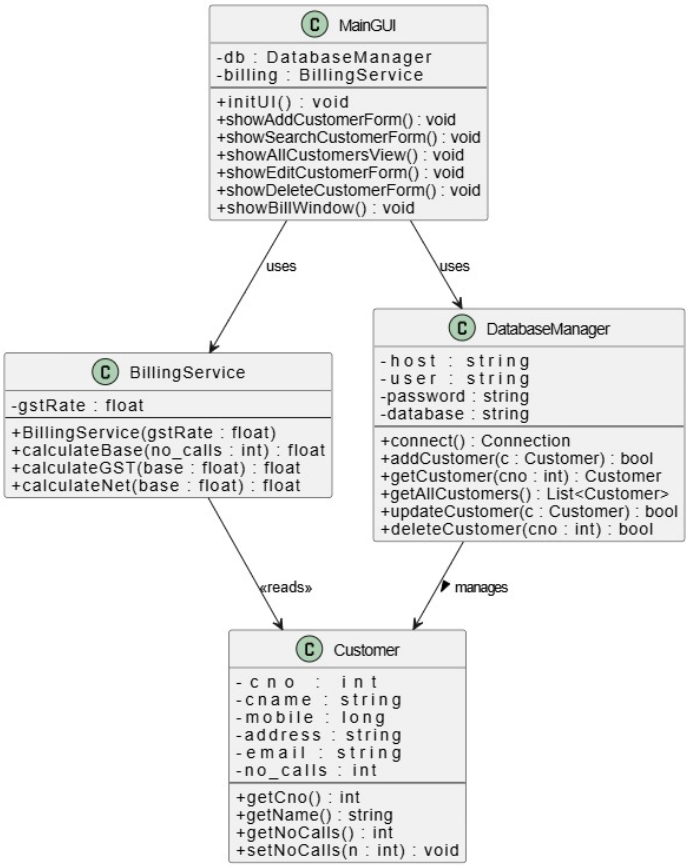
8.1 **Use-Case Diagram**



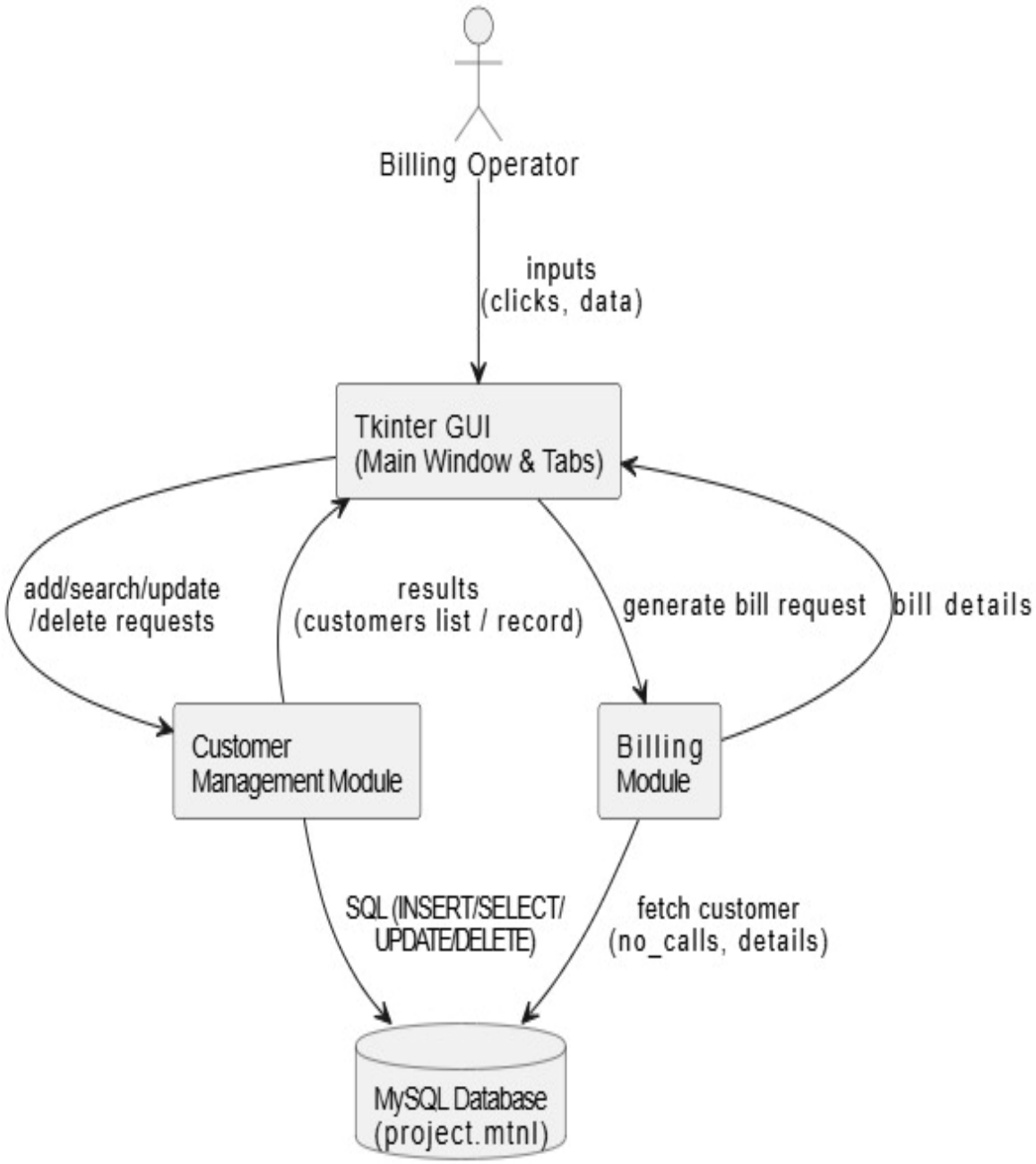
8.2 ER Diagram



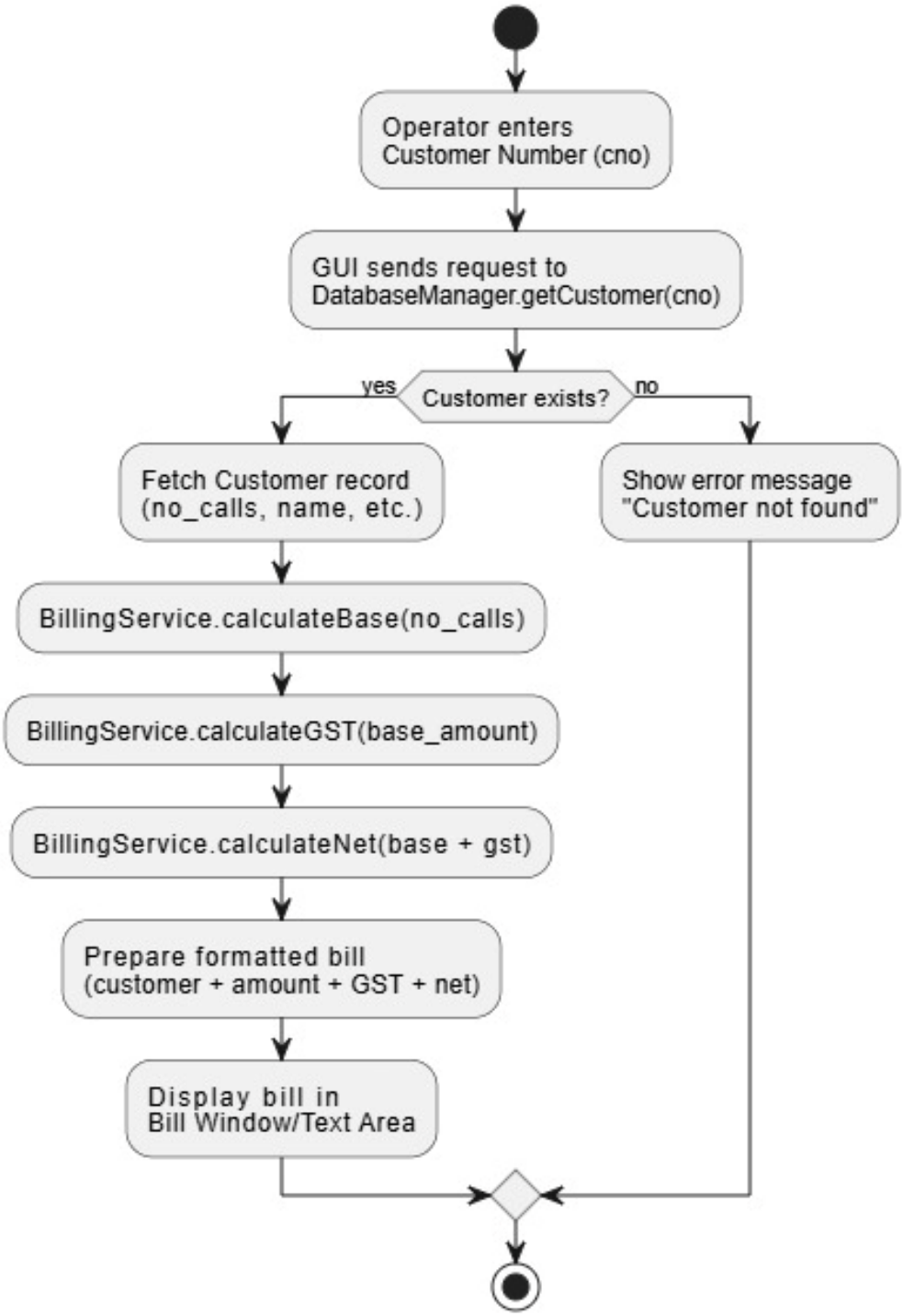
8.3 Class Diagram



8.4 Component (Architecture) Diagram

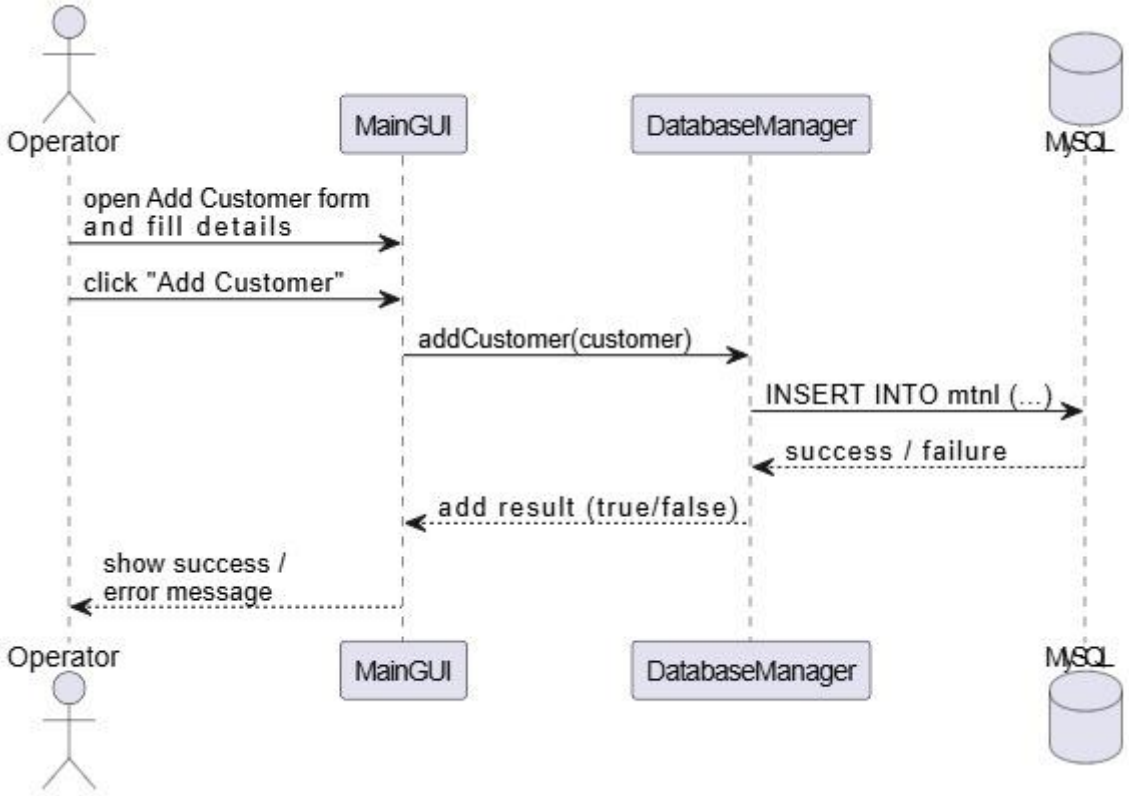


8.5 Activity Diagram

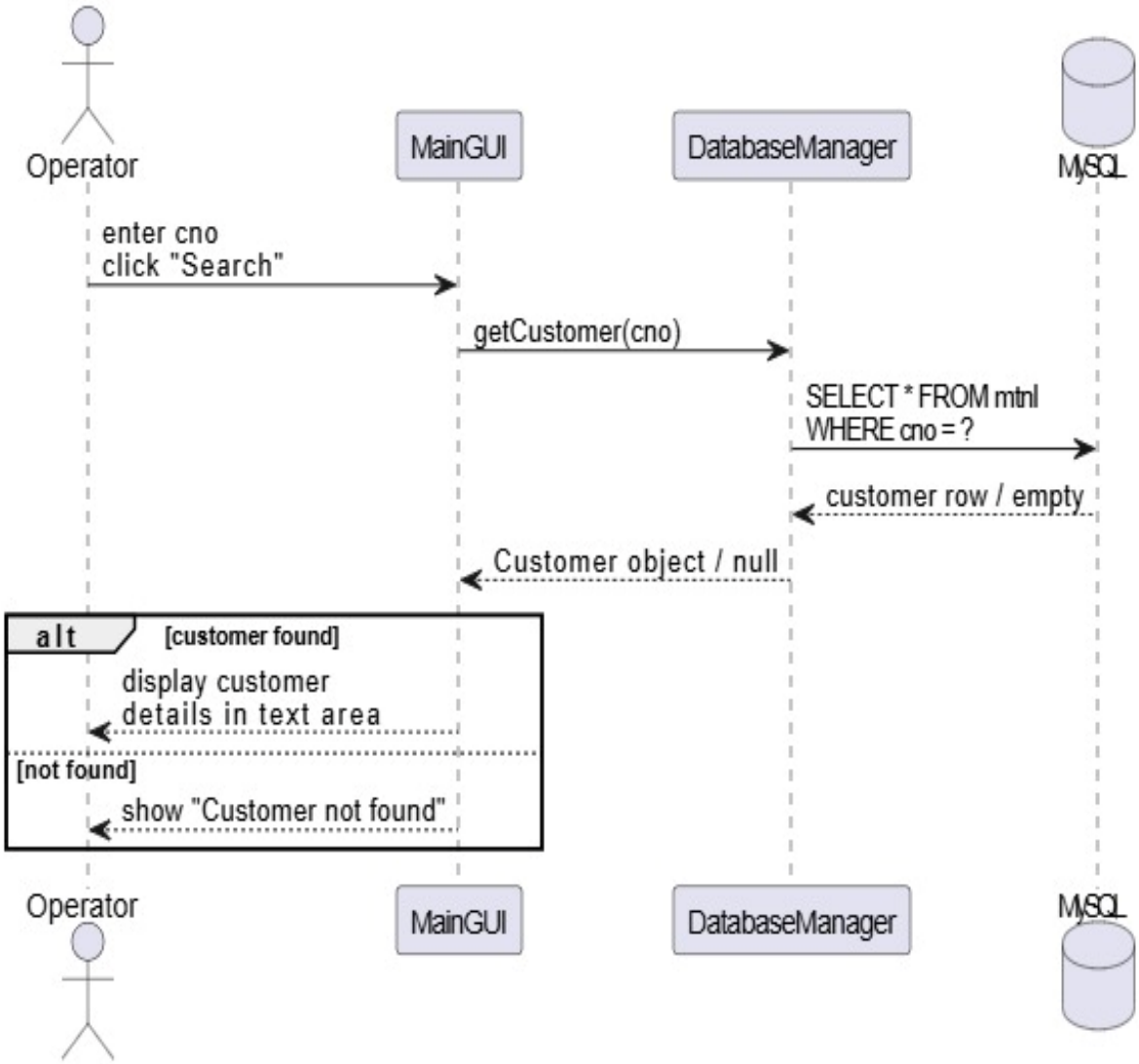

Operator enters
Customer Number (cno)
GUI sends request to
DatabaseManager.getCustomer(cno)
Customer exists?
yes
no
Fetch Customer record
(no_calls, name, etc.)
Show error message
"Customer not found"
BillingService.calculateBase(no_calls)
BillingService.calculateGST(base_amount)
BillingService.calculateNet(base + gst)
Prepare formatted bill
(customer + amount + GST + net)
Display bill in
Bill Window/Text Area

8.6 Sequence Diagrams

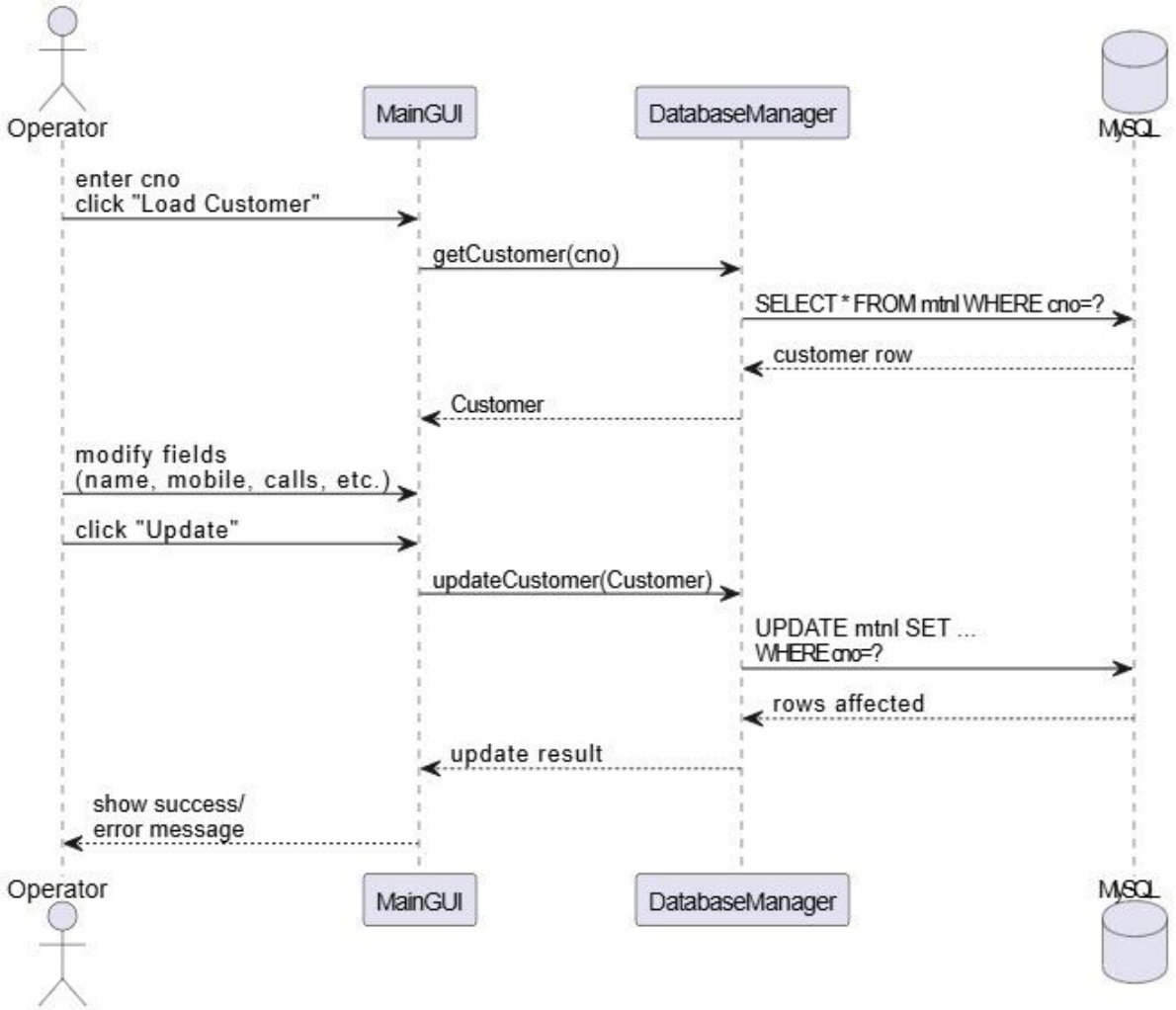
Add Customer Sequence



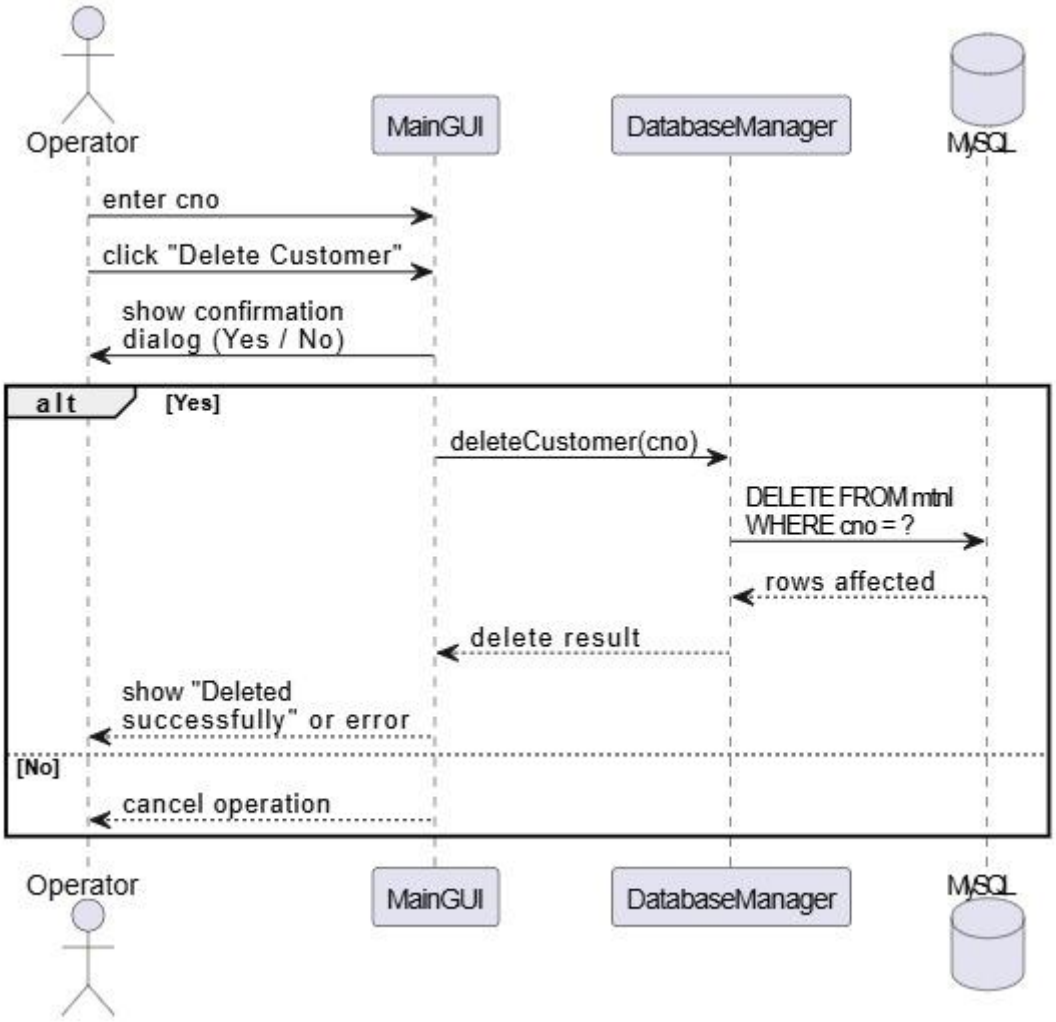
Search Customer Sequence



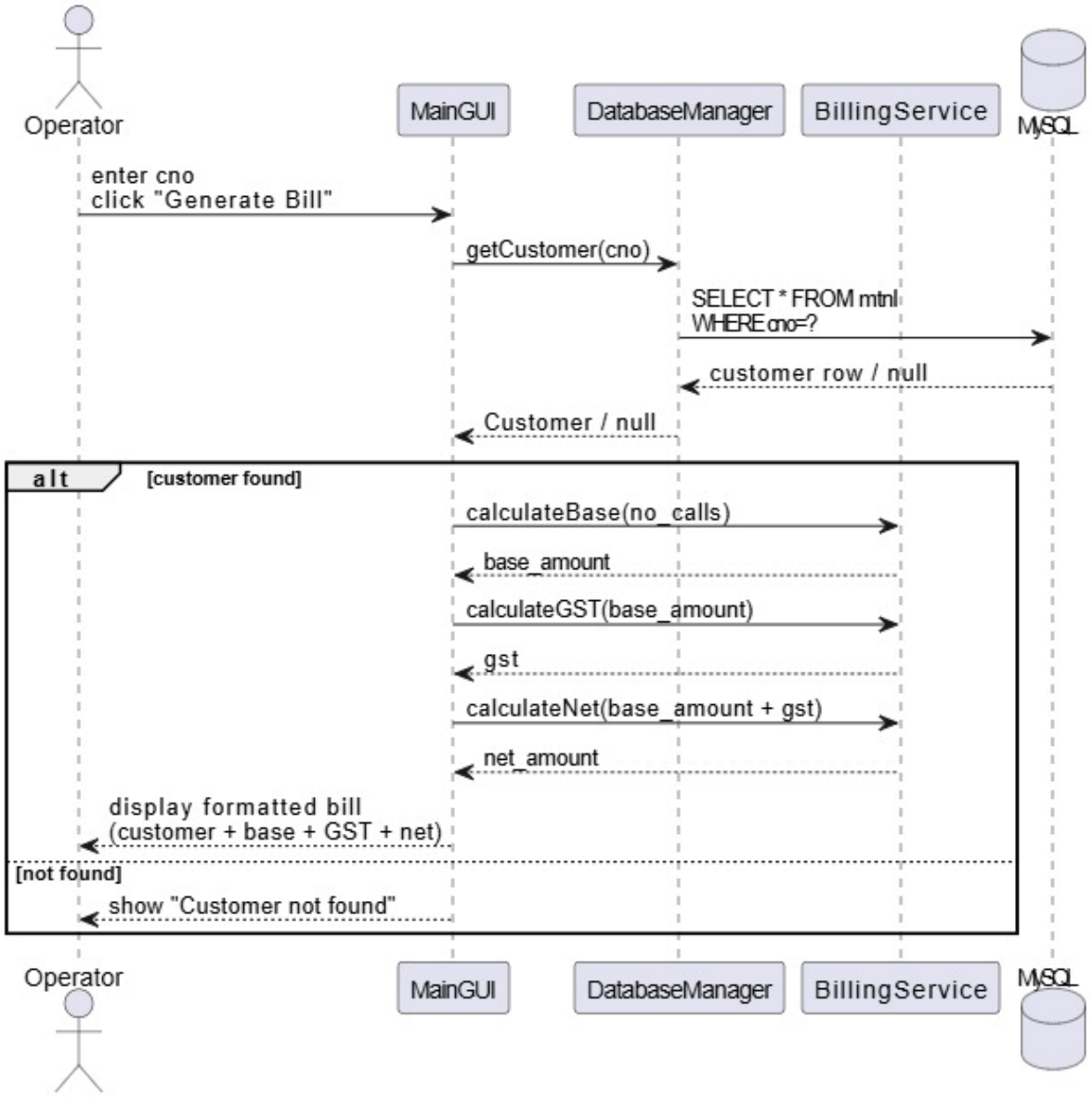
Update Customer Sequence



Delete Customer Sequence

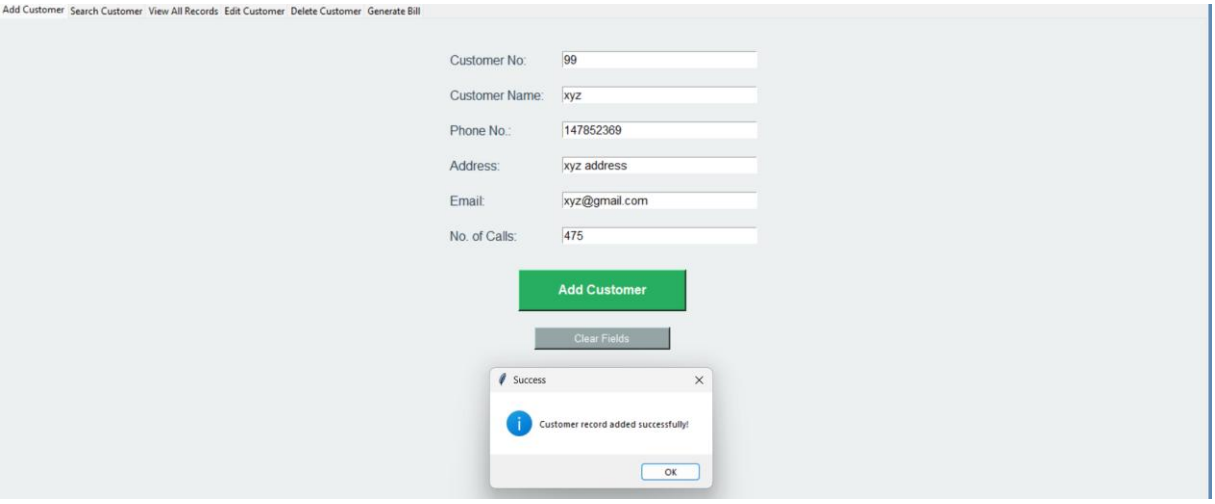


Generate Bill Sequence

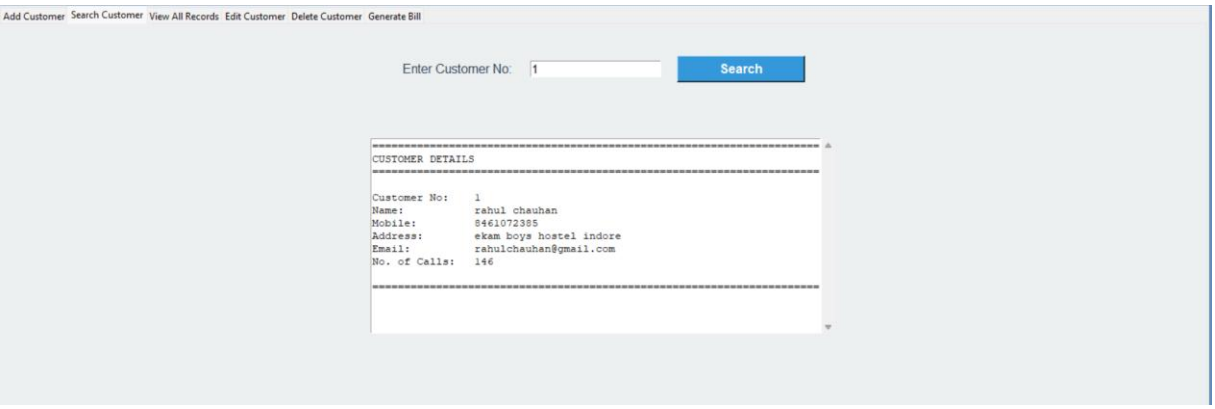


9. User Interface Screens

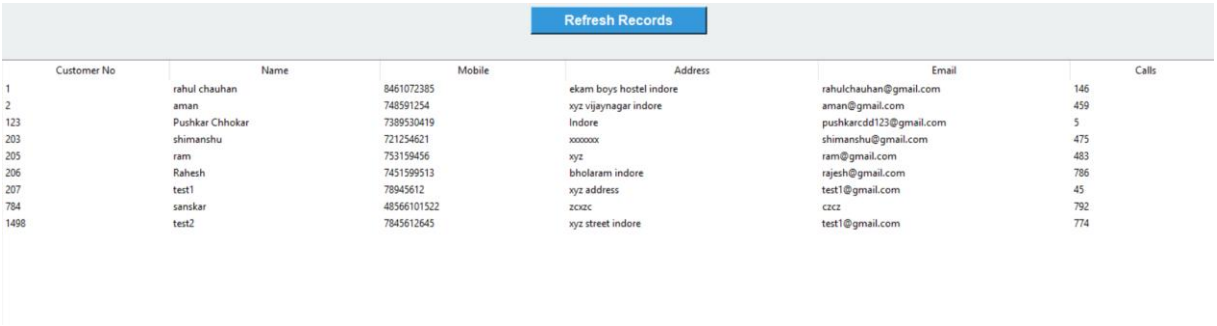
Add Customer



Search Customer

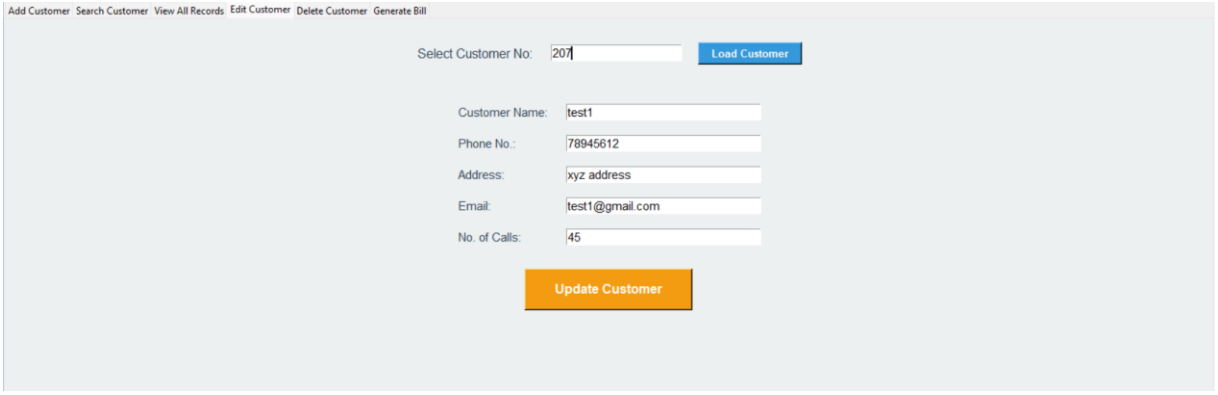


View All Records

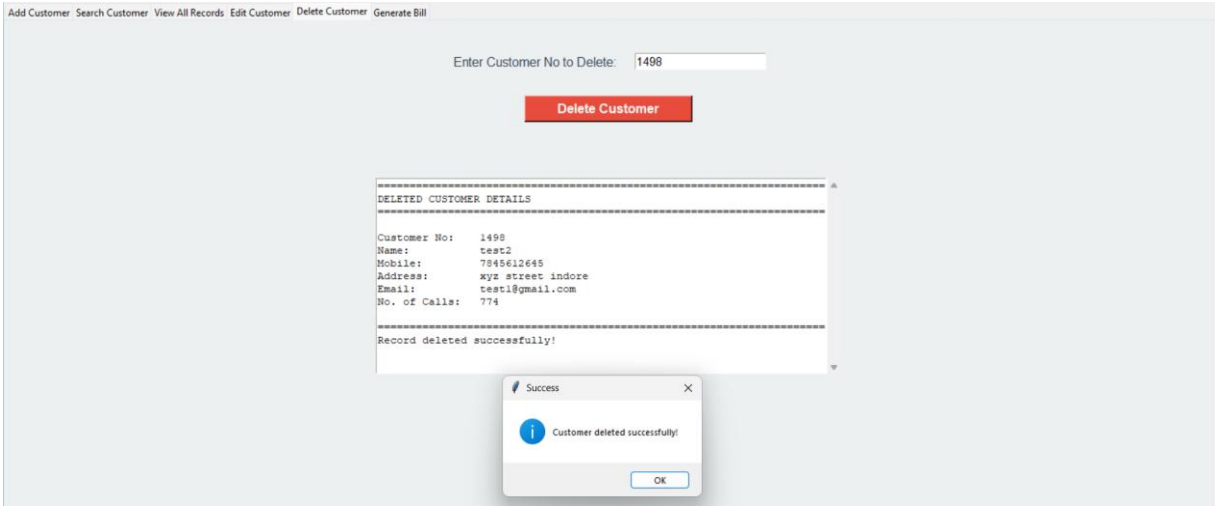


Customer No	Name	Mobile	Address	Email	Calls
1	rahul chauhan	8461072385	ekam boys hostel indore	rahulchauhan@gmail.com	146
2	aman	748591254	xyz vijaynagar indore	aman@gmail.com	459
123	Pushkar Chhokar	7389530419	Indore	pushkarcdd123@gmail.com	5
203	shimanshu	721254621	xxxxxxx	shimanshu@gmail.com	475
205	ram	753159456	xyz	ram@gmail.com	483
206	Rahesh	7451599513	bholaram indore	rajesh@gmail.com	786
207	test1	78945612	xyz address	test1@gmail.com	45
784	sanskar	48566101522	zcxzc	czcz	792
1498	test2	7845612645	xyz street indore	test1@gmail.com	774

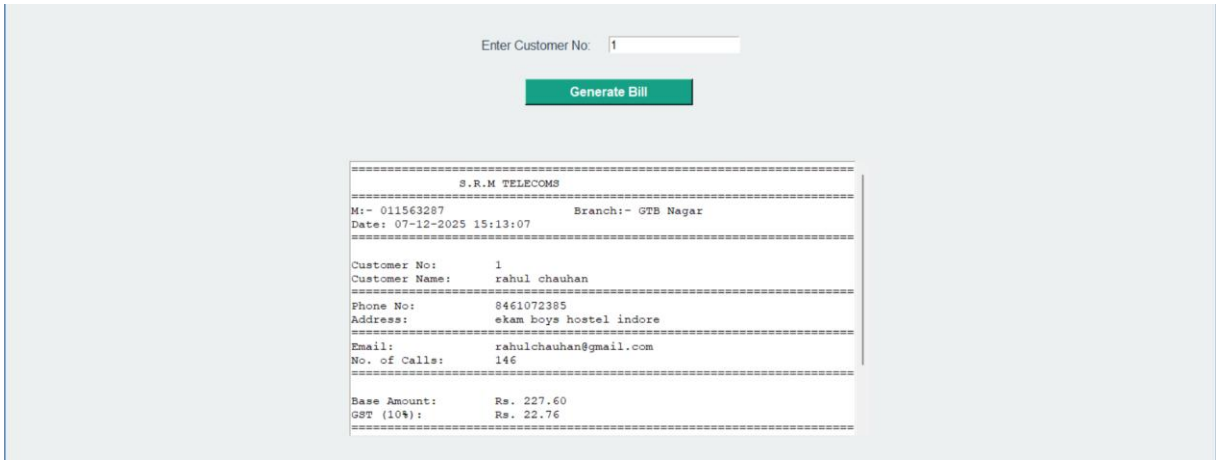
Edit Customer Form



Delete Customer



Generate Bill



10. Results

The Telecom Billing System successfully performs:

- CRUD operations
- Accurate bill generation
- GUI-based interaction
- Database integration

12. Conclusion

This project demonstrates real-world concepts such as database handling, GUI development, and automation. The system reduces manual tasks and enhances billing reliability and consistency.

13. Future Enhancements

- Export bill as PDF
- Add SMS/Email bill delivery
- Cloud deployment
- Multi-user login

14. Bibliography

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